DIVAHARAN D

Embedded Software & Electronics Hardware Design Engineer

Embedded C • STM32 • RTOS • Altium PCB Design • Power Electronics

PROFILE
Embedded and electronics design engineer with around 3 years of hands-on experience spanning firmware development, multilayer PCB design, motor-control electronics, power interfaces, prototyping and hardware validation. Comfortable working across the hardware–firmware boundary, from component selection and schematic/PCB design through bring-up, debugging and test.

WHAT I DO

Capability	Scope
Embedded firmware	Embedded C, STM32, ESP32, FreeRTOS concepts, peripheral drivers, communication interfaces and control logic.
PCB & electronics design	Altium Designer, 4-layer board design, schematic capture, component selection, power distribution and layout reviews.
Motor & power electronics	PMSM/BLDC control interfaces, current/voltage sensing, protection, gate-driver interfaces and brake-chopper control.
Prototype & validation	Board bring-up, lab testing, fault diagnosis, sensor integration, debugging and design iteration.

TECHNICAL TOOLKIT

MCUs: STM32H7, STM32F3, STM32G0, ESP32-C6
Languages: Embedded C, C
Interfaces: CAN, UART, SPI, I²C, RS-485, Ethernet, USB
Design & test: Altium Designer, KiCad, STM32CubeIDE, CubeMX, LTspice, PSpice, MATLAB, oscilloscope and bench power-supply workflows

SELECTED ENGINEERING PROJECTS

01 | STM32H745 Master Controller — 4-layer PCB

Designed a 4-layer controller around STM32H745 for a real-time embedded application. The board integrates dual isolated CAN interfaces, 4G connectivity, flash memory, isolated RS-485 power monitoring, Ethernet using LAN8742-family PHY, USB and USB Type-C connectivity.

Focus: system architecture, schematic and PCB design, interface integration, power-domain planning and firmware bring-up.

02 | 1.2 kW PMSM Motor Controller — STM32F302

Developed a motor-control electronics and firmware platform using STM32F302. Functional elements include Hall-sensor feedback, current sensing, CAN communication and over-current, under-voltage and over-voltage protection, with FreeRTOS-based software structure.

Focus: embedded C, control-oriented firmware, sensor feedback, protection logic and hardware/firmware integration.

03 | Brake Chopper / Regenerative Energy Dissipation

Designed a 4-layer brake-chopper controller using STM32G0B1. Isolated bus-voltage sensing uses TI AMC3311, isolated current sensing uses AMC3302, and the controller uses CAN communication with closed-loop control to manage the braking resistor path.

Validation: prototype tested with a 400 V DC supply and load in a controlled laboratory setup. Detailed performance values should be added only when verified test data is approved for sharing.

04 | Catcher Controller Module

Worked on a controller architecture using STM32G0B1 and BTM9021EP H-bridge devices for two catcher motors. The design combines SPI-based control, encoder feedback, limit-switch inputs and IPS-related sensing for motor actuation.

Focus: motor-driver integration, feedback-based control, I/O protection and board-level debugging.

05 | Isolated Power & CAN Interface

Designed isolated CAN power/interface sections using SN6501 transformer-driver architecture, transformer coupling and TPS76350 regulation, alongside isolated CAN transceiver integration.

Focus: isolation boundary, isolated supply design, communication robustness and PCB layout considerations.

06 | Autonomous EV Pod Electronics — Multi-board platform

Contributed to electronics development for an autonomous electric pod prototype, including controller, power-switch/comparator and catcher-related boards. Work covered component selection, PCB design, sensor and motor interfaces, board fabrication coordination, bring-up and failure analysis.

Project details are summarized at a high level. Share detailed images, schematics or performance figures only where project confidentiality and employer permissions allow.

ENGAGEMENTS & SERVICES

I can support early-stage teams, product companies and engineering groups with focused electronics development work.

PCB design support

Schematic capture, component selection, 2–4 layer PCB layout, design review and fabrication package preparation.

Embedded firmware

STM32/ESP32 firmware modules, peripheral integration, communication drivers and hardware bring-up.

Motor-control electronics

Motor-driver interface design, current/voltage feedback, protection circuits and controller integration.

Prototype debugging

Power-up checks, interface troubleshooting, board fault isolation, rework support and validation planning.

Design documentation

BOM preparation, interface descriptions, test procedures, bring-up notes and engineering handover documents.

WORKING APPROACH

1. Understand requirements and constraints • 2. Define architecture and interfaces • 3. Design and review • 4. Prototype and bring up • 5. Test, document and iterate

CAREER SUMMARY

Hands-on experience in embedded software and electronics hardware development, with practical exposure to multilayer PCB design, STM32-based control systems, isolated sensing and communication, motor-control interfaces and prototype validation. Interested in embedded hardware/firmware roles and project-based engineering collaborations.

CONTACT

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LinkedIn	Add your LinkedIn profile URL
GitHub / Portfolio	Add your GitHub and website URLs
Location	Chennai, Tamil Nadu, India

Note: This portfolio is an initial public-facing draft. Confirm project-sharing permissions, replace contact placeholders, and add approved PCB renders, prototype photos and test graphs before publishing.